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LED lighting

Though the 'LED Lighting For Commercial Applications' article published in June issue gives latest information, it does not make comparison (in terms of lumens) with popular tubelights fitted with electronic choke. Can LED lights replace a 40-watt tubelight?

K.V. Raman
Chennai

The author Srabani Sen replies:

LED lighting is gaining popularity in more and more commercial and industrial applications as the technology is much safer, cheaper and more energy-efficient than other lighting sources. LED lighting fixtures consume less energy, thus inspiring many commercial and industrial applications to convert to LED lighting. Besides, LED lighting is visually appealing too. LED lights can replace CFL or other tubelights. In the technology focus article 'Twinkle, Twinkle Little LEDs,' in this issue, comparisons have been made between wattage, lumen, efficacy and hours between ordinary bulb, CFL and LED.

We will try to cover the details on replacing a 40-watt tubelight with LED in the 'Qs&As' section in one of our forthcoming issues.

iButton Digital Access System

I have following queries in 'Secure Digital Access System using iButton' project published in November 2010 issue:

1. How is the iButton data stored in the microcontroller?
2. How can the same system be used for five different iButtons?
3. With the same code and circuit, will this project work if I used a similar but different iButton purchased from the market?

Santosh Mishra
Kathmandu, Nepal

The author Dr Harshad G. Chitambar replies:

1. iButton data is stored in an array in the code for the microcontroller. Kindly check the code published in the issue.

2. Store the unique ID of the iButtons in different arrays and access them as individual arrays.

3. Using the same code, you need to change the unique ID of the specific iButton in the array to access the iButton.

Correction: Automatic Water Level Controller

In the 'Automatic Water Level Controller For Pump/Submersible Pump' project on page 101 in this very issue, please note following corrections:

In Fig. 2 (page 102) and parts list (page 104), read resistors R11 and R17 as 47-kilo-ohms. All BC547 transistors must be BC547B type.

On page 103, column 3, under the 'submersible pump starter activation circuit' heading, read capacitor C3 as C4; and on page 104, column 1, read R6 as R8.

Both RL1 and RL2 relays must be 30-ampere contact rating to handle initial surge current.

B. Shanker Rao
The author

PIC Series-Based Projects

Most of the projects published in EFY are based on ATME/AVR series. Please publish projects based on PIC series of microcontrollers as well. Also request you to publish a project on Tetris game, preferably based on PIC18F4520 microcontroller.

Saloni Bhatia
GJUS&T, Hissar

EFY is a Mind Opener

The 'dekho to jano' is applicable to engineers and enthusiasts/hobbyists/students. I am working in a government R&D lab. From my experience I can say that even complex tasks are solved with simple ideas. EFY is a good source to get ideas. Indeed, its

a mind opener for the vast and fast changing field of electronics.

Pankaj Srivastava
Facebook

Milohm Meter

In the 'Milohm Meter' circuit published in July issue, you have provided an example: A 0.5mm diameter wire having a length of 10 cm would have a resistance of ~9 milohm. The ~9 milohm figure could not be understood. I calculated with the given example but the answer is not within 0-9 milohm range. Please give step-by-step calculation.

Ajoy Kumar Bandyopadhyay
Kolkata

The author Raju Baddi replies:

Resistance (R) of a conductor of length 'l', cross-sectional area 'a' and resistivity 'ρ' is given as:

$$R = \frac{\rho \times l}{a}$$

In our example of commercial copper, $\rho = 17.2 \times 10^{-8}$ ohm-metre

Length of the wire = 10 cm, i.e.,

$l = 10 \times 10^{-2}$ metre

Diameter (d) of the wire = 0.5 mm = 0.5×10^{-3} metre

i.e., the radius (r) = 0.25×10^{-3} metre

Then the cross-sectional area a would be:

$$\pi \times r^2 = 3.142 \times 0.25 \times 10^{-3} \times 0.25 \times 10^{-3} \text{ metre}^2$$

$$a = 0.196 \times 10^{-6} \text{ metre}^2$$

Now using the values of 'l', 'a' and 'ρ' as obtained above in the expression for 'R' we see,

$$\begin{aligned} R &= \frac{\rho \times l}{a} \\ &= \frac{17.2 \times 10^{-8} \times 10 \times 10^{-2}}{0.196 \times 10^{-6}} \\ &= 8.7755 \times 10^{-3} \approx 9 \text{ m}\Omega \end{aligned}$$

LED Display

I need a simple circuit using 5×7-segment LED display to display a number. Please help.

Sanjay Gupta
Facebook

Gurpreet Bhullar replies: Visit the link <http://attiny.com/dotmatrix.htm>